# APS110H1f2026 - September 25, 2026

September 24, 2026 1:32 PM

- Today again… more/additional compositions
- Properties of Matter(s) - today again (and also for this week upcoming) - Structure-Property Relationship
  - Calculation
  - Nano structure
  - Radius - maybe today

(a) 1 metal only

$\frac{1}{8} \times 8 = 1$

$\frac{1}{2} \times 6 = 3$

$\Rightarrow q = n$

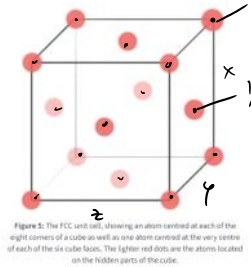


Figure 5: The FCC unit cell, showing an atom centred at each of the eight corners of a cube as well as one atom centred at the very centre of each of the six cube faces. The lighter red dots are the atoms located on the hidden parts of the cube.

Figure 5

(b) density $\rho = \frac{\text{mass of atoms}}{\text{volume of unit cell}} = \frac{g}{cm^3}$ or $\frac{kg}{m^3}$ — you choose which you to us?

$\rho = \frac{n \cdot A}{V_c \cdot N_A}$ ($n$ ← # of atoms in the unit, $A$ ← molar mass, $N_A$ ← Avagadro's)

volume of the unit cell = $x \cdot y \cdot z$

1 nano gram = $1 \times 10^{-9}$ grams.

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(b.1) 2 metals — Pd #1 palladium + Ti #2 titanium

"purity"

$\frac{1}{8} \times 8 = 1$

$\frac{1}{2} \times 6 = 3$

$\rho = \frac{(n^1 \cdot A^1) + (n^2 \cdot A^2)}{V_c \cdot N_a}$

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(b.2) Pd Ti

$\frac{1}{8} \times 8 = 1$

$\frac{1}{2} \times 6 = 3$

inside = 1

Fe iron

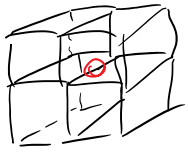
$\rho = \frac{(n^1 \cdot A^1) + (n^2 \cdot A^2) + (n^3 \cdot A^3)}{V_c \cdot N_a}$

"is it pure?" = yes if manufactor

"is it pure?" = No – not whatever manufactor

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(c) ceramic $NaCl = Na^+ + Cl^-$



$\frac{1}{8} \times 8 = 1$

$\frac{1}{4} \times 12 = 3$

$\frac{1}{2} \times 6 = 3$

1 central = 1

$Cl^- = 4$

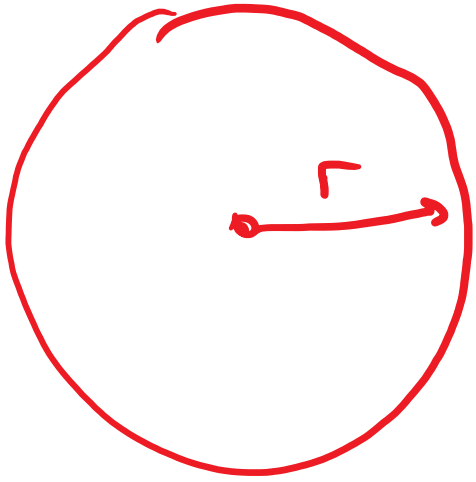
$Na^+ = 4$

$\rho = \frac{(n^1 \cdot A^1) + (n^2 \cdot A^2)}{V_c \cdot N_a}$

Same equation as above.

# Radius - maybe today

September 25, 2026 7:56 AM



# PERIODIC TABLE OF ELEMENTS

